Capstone Team Final Report
Gerontology and Diabetes Research Laboratory (GDRL)
AI Screening of Spontaneous Speech to Detect Alzheimer's

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1 Summary

Alzheimer's disease (AD) is a progressive neurodegenerative disorder which is indicated by cognitive decline and memory loss, leading to impaired daily functioning. Due to the fact that there is still no cure for the disease, early detection of AD is crucial for timely intervention and management of the disease, which would allow the patients to maintain a better quality of life. Apart from the lengthy, expensive and invasive detection techniques like brain imaging and lumbar puncturing, a widely used tool for assessing cognitive function and language abilities in the context of AD is the Cookie Theft picture description task from the Boston Diagnostic Aphasia Examination (BDAE) [1]. In this particular assessment, the participants are shown a picture of the scene where a boy is trying to steal some cookies from the cookie jar while his mother is busy washing the dishes 1. The participants are then told to describe the scene in detail, and a trained professional analyzes the output to find relevant linguistic and cognitive markers that might indicate AD. One prominent approach in the field is to analyze the audio and linguistic data using domain expertise and manually engineer features that can be used for the classification task [2, 3]. This approach, however, misses out on the higher-order features present in the data that are not apparent to the human expert. In recent years, artificial intelligence (AI) and machine learning (ML) techniques have become quite popular due to their ability to find and use higher-order patterns that may be difficult for humans to find. AI models have shown remarkable success in various medical applications, including diagnosing

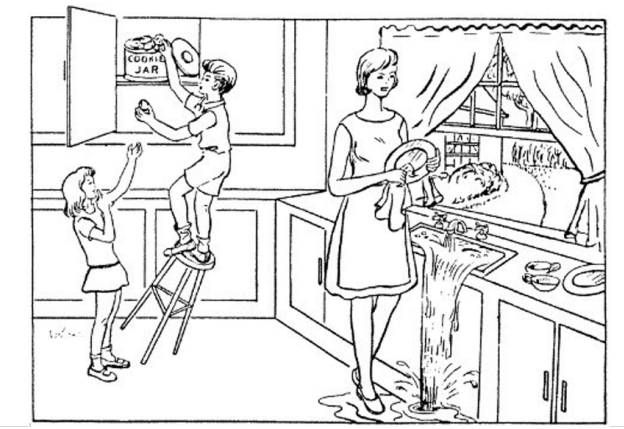


Figure 1: Cookie Theft picture description task from the Boston Diagnostic Aphasia Examination

neurodegenerative diseases [4]. By utilizing the powerful capabilities of these models, our team has developed an automated pipeline to analyze the audio recordings of the people giving the cookie test and then to classify whether an individual has Alzheimer's disease or not. Our pipeline integrates advanced natural language processing (NLP) techniques to extract linguistic features from the audio transcripts. It applies state-of-the-art machine learning algorithms to these features for classification. In this project, we utilized audio recordings from the DementiaBank English Pitt Corpus [5] which contains Cookie Theft picture descriptions from individuals with and without AD. Our primary focus was on the audio transcripts, as they produced the best results in our experiments. We did, however, also explore both audio and hybrid techniques as well to explore the potential of acoustic features in improving the classification results. While our focus is on the task of classification, we also evaluate the results of regression models based on a few approaches. Our main contributions to this project are as follows:

1. We implemented and compared multiple ML algorithms, such as Tfidf word frequency + Logistic regression, Bert text embeddings and summarization, SetFit fine-tuning and audio embeddings.

2. We rigorously evaluated and validated our models using appropriate performance metrics and cross-validation techniques to assess their generalizability and robustness.

3. We are currently working with Dr. Kenneth Madden on deploying this pipeline and testing it on real work data.

2 Related Works

2.1 Alzheimer's Disease and Dementia

Alzheimer's Disease (AD) is a neuro-degenerative disease that is the leading cause of dementia, which results in declines in cognition, behavior, and social skills. Risk factors encompass older age as well as variables such as but not limited to genetic factors and vascular diseases. [6] It is estimated that over 50 million individuals have dementia globally and this number is expected to balloon to over 150 million people by 2050. The AD continuum consists of AD dementia, prodromal AD, and preclinical AD, corresponding to an estimated number of 32, 69, and 315 million people globally. [7] Unfortunately, no cure exists for Alzheimer's disease and there only exists treatments that improve the symptoms of patients, which are more effective when they are administered in early stages. Consequently, it has become imperative for potential patients to be diagnosed earlier on in the process. At this moment in time, while advances in blood biomarkers present exciting opportunities to lower the cost of diagnoses, traditional approaches involve lumbar punctures and amyloid-PET scans that are resource intensive. [8]

2.2 Natural language processing approaches

Due to the cost of traditional diagnostic approaches, many researchers have turned to machine learning models to identify Alzheimer's disease in patients by evaluating changes in patterns of linguistic features using both text and audio data. [9] This is based on the understanding that linguistic ability is one of the first observable factors to deteriorate as a result of Alzheimer's Disease and dementia. Using results of cognitive tests, researchers have employed a variety of methods including text-based lexical features, syntactic features, semantic features, pragmatics, as well as acoustic prosodic features, spectral features, vocal quality, and ASR-related features. In addition, with the rise of deep learning in recent years, researchers have utilized transformer-based models to convert texts and audio data into corresponding representations for the purpose of employing classification and regression approaches. Both the ADReSS and ADReSSo challenges at Interspeech 2020 and 2021 facilitated progress in this field by making standardised Alzheimer's dementia datasets in English available for researchers to build classification and regression models. [10] Cross-lingual analyses have been conducted as well using support vector machines and artificial neural networks for the task of classification based on grammar based linguistic features, word embeddings generated from Bidirectional Encoder Representations from Transformers (BERT), acoustic features, and wav2vec embeddings. [11] Cross-lingual models built using the LIWC feature set and word embeddings with a support vector machine k-nearest neighbor head on top showed that the best results employed deep learning approaches. [12]

3 Data

In this project, we have worked with 2 datasets. The first, and original dataset given for the project is ADReSS. In the later phase of the project, we worked with a relatively larger, dataset called Pitt. Pitt can be considered as the superset of the ADReSS dataset. Through out this report, we will refer the ADReSS dataset as the "original dataset", and Pitt dataset as the "Pitt dataset".

3.1 Description

3.1.1 ADReSS Dataset Description

The ADReSS (Alzheimer's Dementia Recognition through Spontaneous Speech) dataset [13] is created to compare approaches to detecting cognitive impairment and decline using spontaneous speech and their transcripts. This dataset is part of the ADReSS Challenge, which aims to tackle three major prediction tasks: detection of Alzheimer's Disease (AD), inference of cognitive testing scores, and prediction of cognitive decline. The dataset includes speech recordings from participants performing specific tasks to elicit spontaneous speech along with the transcriptions. The ADReSS dataset comprises two primary datasets

AD Classification and MMSE Prediction Dataset:

- **Source:** Speech recordings from individuals diagnosed with Alzheimer's Dementia and cognitively normal controls and their corresponding transcripts.
- **Task:** Participants describe the Cookie Theft picture from the Boston Diagnostic Aphasia Examination.
- **Data Matching:** The dataset is matched for age and gender using a propensity score approach to minimize biases.
- **Segmentation:** Recordings are segmented into vocalization sequences with speaker identifiers.
- **Transcriptions**: Transcripts were annotated using the CHAT coding system [14].
- **Preprocessing:** Acoustic enhancements include stationary noise removal and volume normalization.

Cognitive Decline Prediction Dataset:

- **Source:** Longitudinal cohort study involving AD patients.
- **Task:** At their baseline visit, participants perform a category fluency task (e.g., naming items in a category), and cognitive decline is assessed over a two-year period.
- **Definition of Decline:** A decline is a reduction in MMSE score of 5 or more points between baseline and year-2 visits.
- **Size:** 105 audio recordings, split similarly into training and test sets.

In our project, we mainly focused on the AD Classification task, with some research done on the MMSE Prediction task as well.

Source and Collection: The data for the ADReSS Challenge was sourced from clinical and experimental settings involving Alzheimer's patients and cognitively normal individuals. The speech recordings were collected during diagnostic and monitoring processes within clinical trials and observational studies. Participants were instructed to perform tasks that elicit spontaneous speech, such as describing a picture (Cookie Theft) or performing a verbal fluency task. These tasks highlight differences in speech patterns and cognitive function associated with Alzheimer's Dementia.

3.1.2 Pitt Dataset Description

The Pitt dataset [15], like the ADReSS dataset, is a vital resource for research in Alzheimer's Disease (AD) recognition through speech analysis. Both datasets focus on analyzing spontaneous speech to detect cognitive impairment, offering valuable insights and facilitating the development of automatic detection systems. The Pitt dataset shares several similarities with the ADReSS dataset in terms of its composition and purpose.

Data Composition:

- **Participants:** It too includes patients diagnosed with Alzheimer's Disease and cognitively normal controls.
- **Speech Tasks:** Participants in both datasets performed tasks designed to elicit spontaneous speech, such as picture descriptions (e.g., the Cookie Theft picture) and narrative recalls.
- **Audio Quality:** Variations in audio quality are present in both datasets, with efforts made to normalize and enhance these recordings to ensure consistency.

Source and Collection: The source and collection methods for the Pitt dataset have similarities with those of the ADReSS dataset. The Pitt dataset originates from the DementiaBank, a shared database for the study of dementia. It is part of the larger TalkBank project, which provides a repository of multimedia interactions for research in human communication. Data collection involved participants from clinical settings who underwent various speech elicitation tasks. These tasks were designed to simulate naturalistic speech, capturing a range of cognitive and linguistic abilities.

3.2 Format

Both datasets contains audio files and text transcription files organized in separate folders for 2 categories:

- AD (Alzheimer's Disease) and CN (Cognitively Normal) in the original dataset
- Control (negative) and Dementia (positive) in the Pitt dataset.

3.3 License

The datasets are provided by registration only. The capstone project must register to become a member of DementiaBank to gain access to the datasets. The datasets are for research purpose only, and not to be publicly accessible.

3.4 Size

- The original ADReSS dataset has 166 examples for the train set and 71 examples for the test set.
- The original Pitt dataset has 104 controls, 208 patients with dementia, and 85 participants with unknown diagnoses. The dataset we are given consists of 99 controls and 193 patients with dementia. There is a total of 549 examples.

3.5 Annotations

The dataset has some annotation but it is mainly the transcription of the audio files. Besides there are segmentations of the audio with start and end time markers.

3.6 MMSE

The Mini Mental State Examination (MMSE) is a screening tool to evaluate the cognitive status of adults based on a 30 point system. The regression models employ the MMSE scores as the dependent variable.

3.7 Data Pre-processing

Both the original dataset and Pitt dataset comes with text and audio files. For text data, they are in separate folders according to the label, e.g. AD/CN for the original dataset and Control/Dementia for the Pitt dataset. Each folder contains separate text files (.txt for the original dataset and .cha for the Pitt dataset).

We did text data pre-processing to read the text files, remove unwanted information such as white spaces at the beginning and end of files. For the Pitt dataset, we only extracted the transcription lines by the participants (with PAR tag) and ignored text transcription by the investigators (with INV tag).

Then we combined the text into a csv file, with the file_id, text and corresponding labels (AD/CN) for next steps in the pipeline.

3.8 Statistics

3.8.1 Original dataset

The original train dataset is relatively balanced with 87/166 (52%) examples having the AD label and 79/166 (48%) examples having the CN label.

We have divided the original dataset into 2 equal subsets for training and development respectively. Each subset contains 83 examples. The splits are done randomly using Pandas sample function with fraction of 0.5 and a specific random seed number (42).

Top frequent words in the original dataset and their counts:

- AD group: [('picture', 102), ('going', 100), ('water', 98), ('okay', 80), ('tell', 69), ('cookie', 63), ('cookie jar', 59), ('see', 58), ('boy', 57), ('anything', 56), ('everything', 52), ('dishes', 52), ('mother', 46), ('sink', 45), ('look', 44), ('see going', 42), ('girl', 41), ('oh', 41), ('one', 38), ('floor', 36), ('well', 36), ('running', 33), ('happening', 32), ('kid', 30), ('hand', 28), ('right', 27), ('know', 26), ('getting', 25), ('yeah', 24), ('want', 24), ('stool', 24), ('thing', 23), ('see happening', 23), ('trying', 22), ('fall', 22), ('something', 21), ('window', 21), ('think', 21), ('washing dishes', 21), ('falling', 20), ('say', 18), ('now', 18), ('plate', 17), ('action', 17), ('little girl', 17), ('okay good', 17), ('still', 16), ('go', 15), ('washing', 15), ('chair', 15)]
- CN group: [('water', 102), ('cookie', 97), ('sink', 85), ('mother', 75), ('cookie jar', 75), ('boy', 71), ('window', 69), ('stool', 54), ('see', 53), ('picture', 53), ('going', 53), ('one', 52), ('look', 52), ('dishes', 51), ('girl', 46), ('running', 46), ('everything', 45), ('okay', 41), ('reaching', 40), ('getting', 38), ('tell', 38), ('floor', 37), ('hand', 35), ('drying', 34), ('overflowing', 33), ('right', 32), ('curtain', 31), ('standing', 30), ('open', 29), ('falling', 28), ('fall', 26), ('sister', 26), ('want', 24), ('little girl', 24), ('action', 23), ('plate', 22), ('anything', 21), ('see going', 21), ('know', 20), ('looking', 20), ('kitchen', 20), ('trying', 20), ('outside', 19), ('dish', 19), ('another', 19), ('store', 19), ('washing dishes', 18), ('onto', 17), ('thing', 17), ('little boy', 17)]

The summary statistics for MMSE scores are presented in Table 1:

Target	Count	Min	Mean	Median	Max
AD	87	3	17.4	18.0	28.0
CN	79	26	29.0	29.0	30.0

Table 1: Summary statistics for MMSE scores for targets AD and CN in Original Dataset

3.8.2 Pitt dataset

The Pitt dataset has a total of 549 examples, with about 292 unique person IDs. Each person may have one or more associated examples with them taken in different dates.

We have divided the Pitt dataset into 3 subsets:

- The train set: contains 337 examples

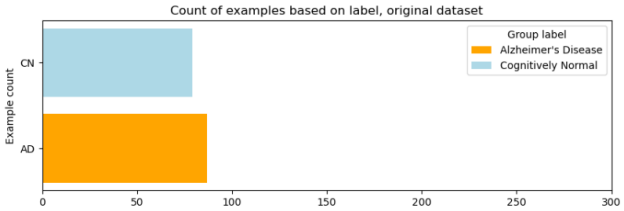


Figure 2: Example counts in original dataset

- The development set: contains 115 examples
- The test set: contains 97 examples

In the Pitt train set, there are 337 examples, in which the counts for AD and CN groups are 187 (55%) and 150 (45%) respectively.

Top frequent words in the Pitt train set and their counts:

- AD group: [('uh', 430), ('cookie', 177), ('water', 175), ('sink', 161), ('cookie jar', 143), ('stool', 134), ('boy', 128), ('dishe', 127), ('well', 123), ('mother', 123), ('oh', 102), ('see', 99), ('laughs', 93), ('floor', 89), ('girl', 86), ('window', 79), ('one', 76), ('know', 72), ('um', 68), ('g', 67), ('little girl', 64), ('look', 62), ('little boy', 60), ('got', 56), ('hand', 52), ('kid', 51), ('guess', 48), ('going', 48), ('xxx', 47), ('sister', 47), ('think', 46), ('dryin g', 45), ('okay', 44), ('standing', 44), ('reaching', 43), ('g dishes', 43), ('runnin g', 43), ('falling', 41), ('gonna', 41), ('open', 39), ('dish', 37), ('s', 36), ('want', 36), ('something', 35), ('getting', 35), ('running', 33), ('drying', 33), ('yeah', 33), ('washin g', 33), ('plate', 32)]
- CN group: [('uh', 345), ('mother', 176), ('water', 145), ('cookie', 145), ('stool', 144), ('sink', 139), ('um', 111), ('boy', 100), ('dishes', 99), ('see', 97), ('window', 92), ('cookie jar', 86), ('okay', 80), ('action', 73), ('standing', 70), ('little girl', 69), ('reaching', 68), ('floor', 62), ('girl', 60), ('look', 59), ('g', 57), ('drying', 57), ('hand', 57), ('one', 53), ('curtain', 53), ('well', 49), ('open', 48), ('little boy', 48), ('running', 47), ('falling', 47), ('outside', 45), ('going', 44), ('overflowing', 44), ('sister', 39), ('oh', 38), ('plate', 37), ('know', 37), ('getting', 36), ('fall', 35), ('out_of', 35), ('drying dishes', 33), ('want', 32), ('blowing', 29), ('dish', 29), ('s', 27), ('guess', 27), ('kitchen', 25), ('xxx', 25), ('laugh', 25), ('washing dishes', 24)]

The summary statistics for MMSE scores of the Pitt dataset for each test are shown in Table 2 but please note that certain tests did not have a MMSE score associated with them:

Target	Count	Min	Mean	Median	Max
AD	276	1	19.7	20.0	30.0
CN	182	24	29.1	29.5	30.0

Table 2: Summary statistics for MMSE scores for targets AD and CN in the Pitt dataset

3.9 Visualizations

3.9.1 Original dataset

Counts of examples based on target label (AD/CN) in the original dataset:

We also have word-clouds for the original dataset, for AD group 3 and CN group 3. In which, the most frequent words in the text (transcription) are highlighted, with more frequent words have bigger font size.

3.9.2 Pitt dataset

Count of examples based on target labels (AD/CN) for the Pitt train dataset 5:

We also have word-clouds for the Pitt train dataset, for AD group 6 and CN group 7. In which, the most frequent words in the text (transcription) are highlighted, with more frequent words have bigger font size

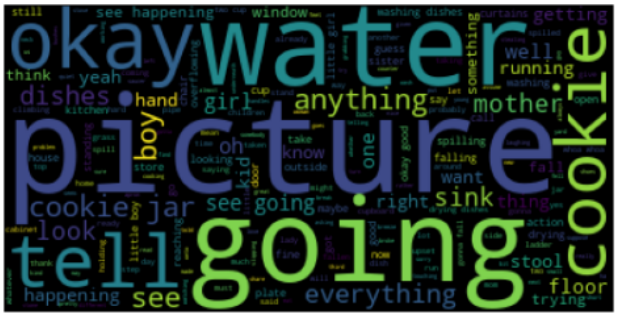


Figure 3: Word-cloud for AD group, original dataset

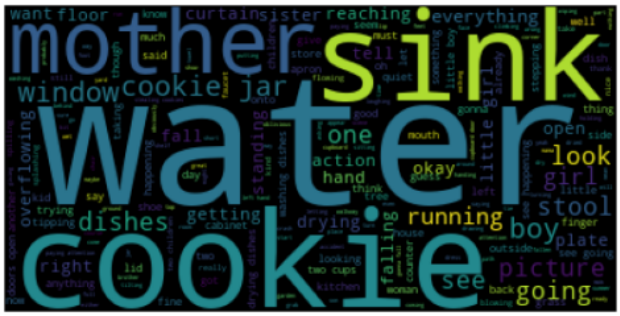


Figure 4: Word-cloud for CN group, original dataset

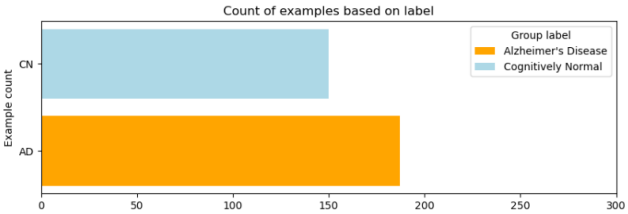


Figure 5: Count of examples in Pitt train set



Figure 6: Word-cloud for AD group, Pitt train set



Figure 7: Word-cloud for CN group, Pitt train set

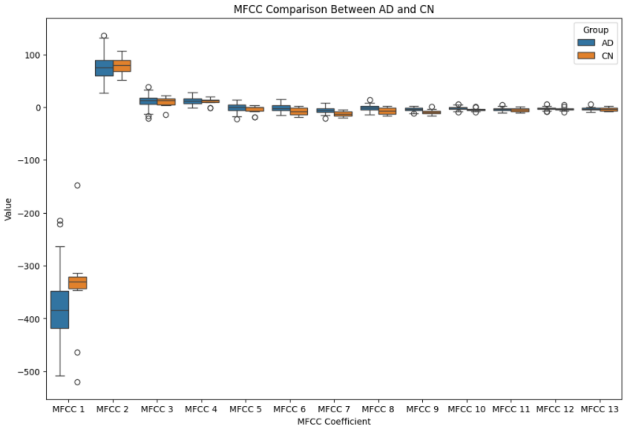


Figure 8: MFCC comparison AD vs CN

3.10 Audio EDA

The box plots provide a visual comparison of various audio features between individuals with Alzheimer's disease (AD) and cognitively normal (CN) individuals. The first plot, which compares Mel-frequency cepstral coefficients (MFCCs), shows notable differences between the groups in the first few coefficients. Specifically, MFCC 1 and MFCC 2 exhibit significant variation between AD and CN, indicating that these coefficients could be important markers in distinguishing between the two groups. The larger spread and higher variability in the MFCC values for the AD group suggest more inconsistency in the speech patterns of those with Alzheimer's.

The Chroma feature comparison reveals less distinct separation between AD and CN groups, though there are still observable differences. In particular, the AD group tends to have higher mean values in several chroma features compared to the CN group. The Spectral Contrast and Formant comparisons also show differences, with the AD group exhibiting higher spectral contrast values and differences in formant frequencies. These variations suggest that spectral features and formant characteristics could also be useful in differentiating between speech from AD and CN individuals. Overall, the plots indicate that specific audio features, such as MFCCs and spectral contrast, are potentially valuable features to solve the classification problem.

3.11 Audio Overlapping

We created an algorithm in order to calculate the similarity between audio files using two distinct methods: Dynamic Time Warping (DTW) and WeSpeaker. DTW is employed to measure the temporal alignment between two audio sequences, effectively capturing variations in timing and speed by calculating the minimum distance path through a cost matrix representing the differences between the sequences. WeSpeaker, on the

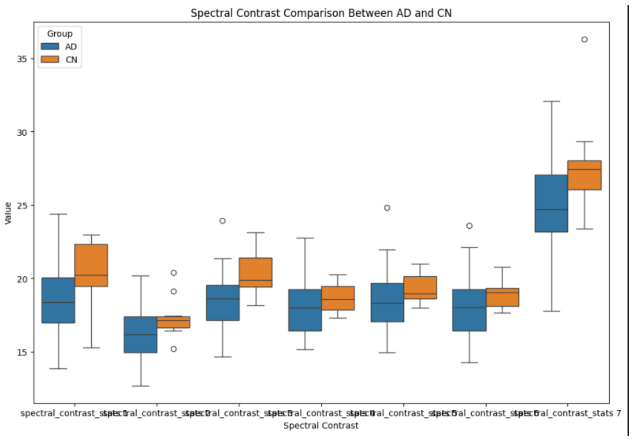


Figure 9: Spectral comparison AD vs CN

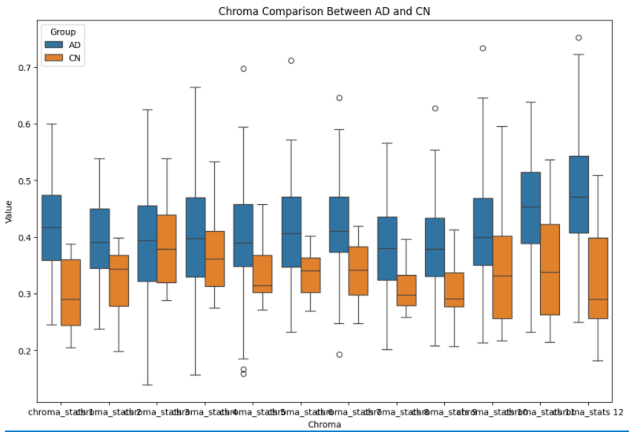


Figure 10: Chroma comparison AD vs CN

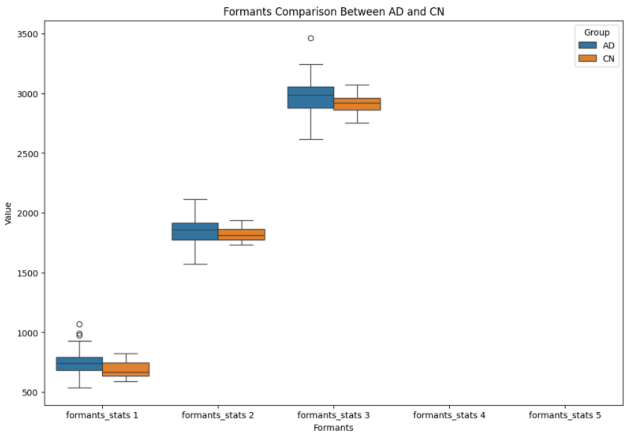


Figure 11: Formats comparison AD vs CN

Method name	Brief description
Classical baseline (Tfidf)	Word frequency and Logisic regression
Hybrid (Tfidf + Text embeddings)	Word frequency + text embdings and Logistic regression
Neural classification	Text embedding and Neural classification
Audio embeddings	Audio embeddings and Logistic regression
Zero- and few-shot LLM	Using Large Language Models to classify text
Setfit	specialized fine-tuning of sentence transformer models on limited data

Table 3: List of classification approaches for original dataset

Method name	Brief description
Classical baseline (Tfidf)	Word frequency and Logisic regression
Hybrid (Tfidf + Text embeddings)	Word frequency + text embdings and Logistic regression
Neural classifier	Text embeddings + neural classifier

Table 4: List of classification approaches for Pitt dataset

other hand, utilizes a pre-trained neural network model to compute speaker embeddings, which are then compared to determine the similarity score between the audio files. By combining these two approaches, the system leverages both time-domain alignment and advanced feature extraction to provide robust similarity measurements, facilitating detailed analysis of speech patterns in different audio datasets.

This approach, however, proved to be inefficient due to its high resource consumption in terms of both cost and time. The DTW method, while effective, is computationally intensive, especially when processing large audio files. Similarly, the WeSpeaker model required significant computational power for embedding extraction and similarity calculation. After consulting with the domain expert, we concluded that the resource expenditure was not justified by the benefits gained.

4 Methods

In this project, we have experimented with many methods to see which approaches produced the best results. The list of approaches covered are summarized in the following table. Each method will be discussed in more details in the subsequent subsections.

For classification task, with original dataset 3:

For classification task, Pitt dataset 4:

For regression task, original dataset 5:

4.1 Approaches

In this section we discuss what approaches we have tried to solve the task. There are many approaches including classical Tfidf + Logistic regression, linear regression, Bert text embeddings, and audio processing pipelines. They produced varying results, and we would select a baseline model and the best-performing model for the final report.

Method name	Brief description
Classical baseline (Tfidf)	Word-frequency + Ridge regression model
Hybrid (Tfidf + Text Embeddings)	Summary + Word-frequency + Ridge regression model

Table 5: List of regression approaches for the original and Pitt dataset

4.1.1 (Classical Baseline) Tfidf word frequency + Logistic regression

When we did the EDA (Exploratory Data Analysis) on the original dataset, we observed that CN group (people without Alzheimer's) tends to describe the cookie thief picture with more details about the children stealing cookies, the stool is about to fall, the mother washing dishes, and water overflowing from the sink. However, people in AD group (people with Alzheimer's) tends to describe the same picture with much less details, using generic and filler words ("uhm", "somebody", etc). Therefore the keywords might play an important roles in determining a piece of text are from people with or without Alzheimer's.

Because of this reason, we choose to use Tfidf vectorizer to convert the text transcription into a fixed-length vector (or list of numbers) that represent the frequency of the words in the text, comparing to the overall set of all text transcriptions (the corpus). Tfidf stands for "Term frequency, Inverse Document Frequency) and is one of the most commonly used method used for feature extraction [16] . The Tfidf vectorizer will reward more weights to words that are relatively rare and more representative to the document, and lower weights to words that are common, and therefore less representative to the text [16]

The Tfidf weight value for a word t in document d is defined as[16]:

$$w_{t,d} = tf_{t,d} \times log_{10}\frac{N}{df_t}$$

where:

- $tf_{t,d}$ is the frequency of the word t in the document d
- N is the number of documents in the corpus
- df_t is the number of documents contains the word t

The text (transcripts) length can be varying between examples in the dataset, however, after using Sklearn Tfidf vectorizer, we can convert them to vectors of same length, for example, 10,000 numbers representing which words are most representative for each example. After converting the text to vectors or list of numbers, we can train the classification and regression models using these fixed-length vectors.

For baseline models, we used a commonly used machine learning library called Sk-learn (also called scikit-learn) for Tfidf processing, logistic regression classification, and ridge regression.

The classical baseline classification model use Tfidf Vectorizer and Sklearn Logistic regression classifier with default settings. On the other hand, the baseline classical regression model uses the same vectorizer and a ridge regression with the evaluation metrics of root mean squared error. The classifier learns from the numerical representative vectors and corresponding label (AD or CN) and will predict labels for text examples in the development and test set, as illustrated in this diagram 12. Similarly, the regression model predicts MMSE scores based on the TFIDF vectors. The classifier output then will be evaluated with accuracy and recall scores.

For our project, because the dataset provided with 2 different labels AD (Alzheimer's Disease) and CN (Cognitively Normal), so we used multi-class Logistic regression classifier instead of binary classifier. It will predict a particular example to be in class AD or CN. Essentially, it works similarly to a binary classification. In the future works, if there are more than 2 labels, for example "Mild AD", "Severe AD", etc, the model can handle these new classes without the need to modify the code heavily.

The accuracy score is the overall percentage of how many correct predictions in all predictions, for all labels (AD/CN).

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions}$$

Recall is defined as[17]

$$Recall = \frac{True\ Positives}{True\ Positive + False\ Negatives}$$

The recall score for AD label is also the percentage of how many correct positive AD predictions in all actual AD labels. We choose to use recall score because it's generally important in medical field that if a number of people actually has a diseases (in our task, it's Alzheimer's Disease), the model should detect correctly as many as possible in this group. The cost of false negative (a person has the disease, but the

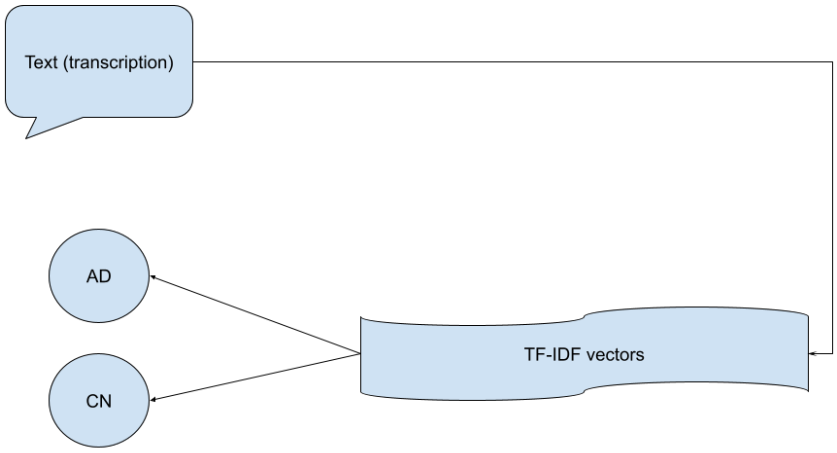


Figure 12: Classical Tfidf model architecture

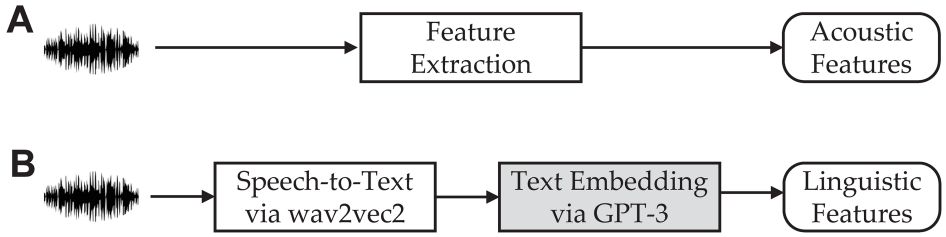


Figure 13: Using GPT-3 to extract text embeddings [18]

model predicted as not having it) can be very high in terms of health and emotion implications, so we would want to measure the AD recall rate, alongside overall accuracy rate.

The numerical results for evaluation for classical method is discussed in depth in the **Evaluation** section.

4.1.2 (Joint-hybrid) Tfidf word frequency + Bert text embeddings + Bert summarization + Logistic regression

Although Tfidf method works well at detecting important keywords in a piece of text, it is not designed to capture the complex semantic meanings of a text. In the cookie thief test, participants are asked to describe what they observe from a particular drawing picture, in which there are several important details such as children stealing cookies, the chair is falling, mother washing dishes and water overflowing from the sink, etc. Although these details can be capture relatively effectively in our baseline models, it might be good to capture finer details of the (text) documents and classify the documents into their corresponding labels (AD/CN).

One of the recent methods to extract semantic meaning of a document is using neural models to produce text embeddings. Agbavor et al. (2022)[18] used GPT-3, one of the popular Large Language Models to calculate text embeddings for the same task (classification and regression for people with or without Alzheimer's Disease) and achieved a relatively remarkable result on the test set with accuracy 0.803 and recall 0.971. The process is illustrated in this illustration13

Because GPT-3 is a closed source model running on the cloud and updated regularly, it could be difficult to replicate the results. In addition, it also might have privacy issues because we would need to send the text over to an external server to process the text and get back the text embeddings. In medical context, it is not desirable to do so. Most institution would prefer to have models running in a self-contained environment to protect data privacy.

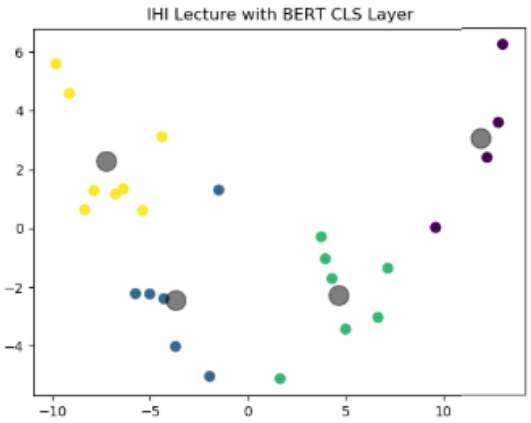


Figure 14: An illustration of clustering sentence in a lecture text [19]

We aim to extend on that research with a more efficient system that is based on open embedding models, and possible to download and run in a self-contained environment.

The brief overview of our hybrid (joint) approach is:

- Pre-process text (transcript) by lower-case it and strip of unwanted characters at the beginning and end.
- Do a automated summarization by using Bert Extractive Summarizer to get a condensed meaning of the text, and add the summary to the end of the text
- Convert the text + summary to fixed-length sparse (long) vector (v1) using Tfidf
- Convert the text + summary to fixed-length dense (short) vector (v2) using Sentence embedings LaBSe model
- Concatenate these 2 vectors into a longer vector (v3)
- Do logistic regression classification on the concatenated vectors for the classification task and ridge regression on the regression task (v3)

Our insight to utilize automatic summarization is based on the observation of the data. Because the speech is spontaneous, some of the words or phrases can be repetitive or not concise. For most people in CN (Cognitively Normal) group, if we summarize up, they would typically talk about "children stealing cookies", "mother washing dishes", "the chair is tiling / falling", "water is overflowing". For people in AD (Alzheimer's Disease) group, if we summarize it, people typically don't mention all these elements, and sometimes talk about something not relevant, or too general to carry significant meaning. Therefore, if we can condense the text by using automatic summarization, it might be useful feature for classification.

Bert Extractive Summarizer [19] is a library to do extractive summarizations using BERT embedding and k-means clustering to find the sentence and phrases that are closest to the summary centroid. In our task, it can reduce the transcription to a few short sentences that capture the condense meaning or main points of the text, and remove repetitive or non-significant words like filler words ("ah", "uhm", etc).

Sentence-transformer (or SBERT)[20] is popular Python library that based on BERT and RoBERTa to produce fixed-length numerical representation vectors of semantic meaning. The vectors can be compared using cosine similarity to find sentences with similar meaning or for classification.

There are many pre-trained models to use with SBERT, and we chose LaBSe (Language-agnostic BERT Sentence Embedding) [21], a multi-lingual embedding model by Google AI that based on BERT and trained with 6 billion translation pairs in 109 languages[22]. Using SBERT (sentence-transformer) library with LaBSe model is simple, efficient and straight forward. It takes in a short document and returns a dense vector (list of numbers) of length 768.

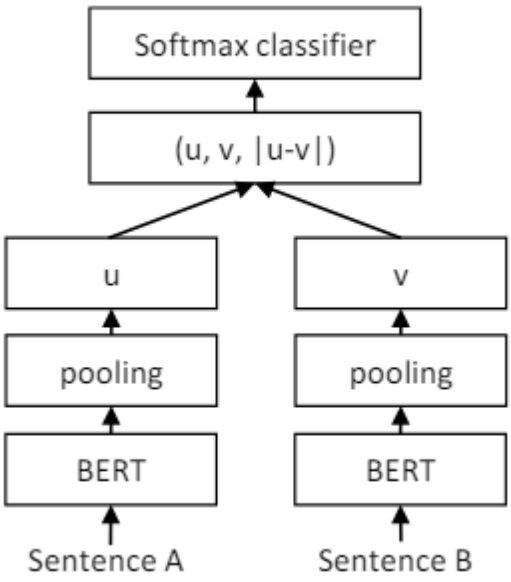


Figure 15: SBERT architecture [20]

When all the components are arranged together in a specific order as in the following diagram 16, we can do logistic regression classification training and prediction on the concatenated vectors representing the text and their summary using Scikit-Learn Logistic Regression tool.

The hybrid approach work produced better accuracy and more balanced (between overall accuracy and AD recall rate) results than the classical baseline models.

4.1.3 Neural classification

One of the less successful model architecture we have tried is neural classification. It has 2 main stages, as in the illustration 17:

- First, we convert text (transcription) into fixed-length vectors using SBERT (sentence transformer) with LaBSe embeding model as described in the hybrid model subsection.
- Then we classify the text into AD/CN label using Sk-learn MLP (Multi-layer Perceptron) neural classifier with the following parameters :
  - solver="lbfgs"
  - alpha=1e-5
  - hidden_layer_sizes=(500, 500, 500)
  - learning_rate_init=0.0001
  - max_iter=500
  - random_state=42

The neural classification model has medium performance, with accuracy and recall on the test set for the original dataset are 0.70 and 0.71 respectively. For the Pitt dataset, the test scores for accuracy and AD recall are 0.72 and 0.85 respectively.

The performance of the neural classifier model is discussed in more details in the **Evaluation** section.

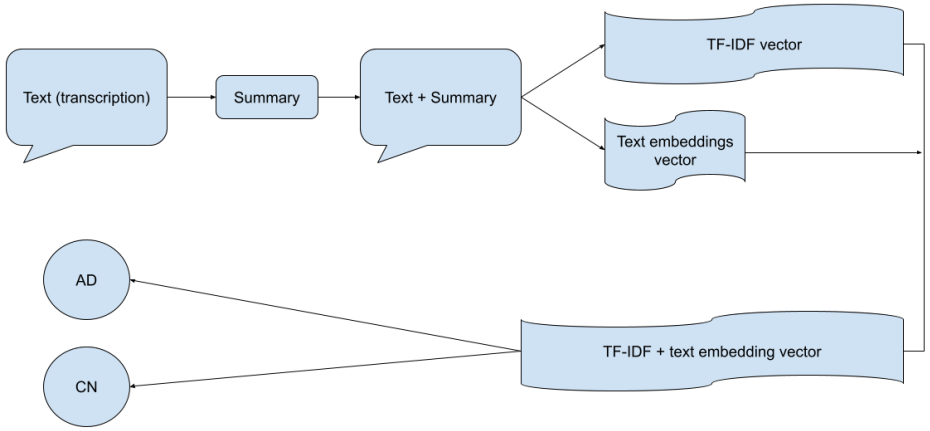


Figure 16: Overview of Hybrid architecture

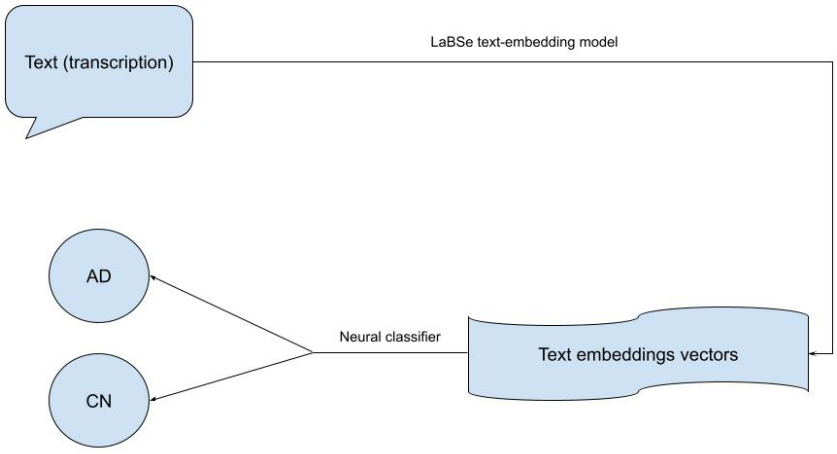


Figure 17: Neural Classification architecture

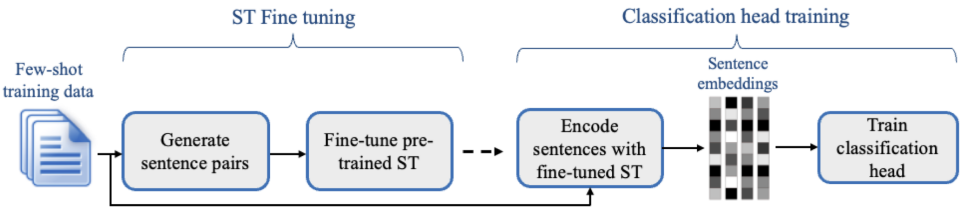


Figure 18: SETFIT 's fine-tuning and training block diagram. [23]

4.1.4 Custom transcriptions + Transformer Embeddings

The initial dataset we had already came with the transcriptions of the audio data. Once we manually evaluated the transcriptions, it was evident that they were generated using an AI model and not manually transcribed. Based on this observation, we wanted to see if the quality of transcriptions had any significant effects on the classification task. For this experiment, we used the Whisper model released by OpenAI. Whisper has multiple variants that range quite a bit in size. We used all of its variants to get the transcriptions and then used a transformer model to get the embeddings of the transcriptions. We then used these embeddings to train a machine-learning model and compare their performance. The results that we got were a bit unexpected. The accuracy and F1 Score of the ML model trained on the embeddings of the original transcriptions were around 75% and 0.75 respectively. We observed improvements in both metrics while using the transcriptions from the whisper model, but the small-en (English version) variant performed better than the bigger models such as medium-en and large. Furthermore, while comparing the results of ML models trained on transcriptions from the medium-en and large whisper variants, the medium-en had higher scores in both metrics. Our hypothesis for this observation is that better models may correct some mistakes like stuttering and mispronunciations that are necessary for the classification task.

4.1.5 SetFit for Efficient Few-Shot Learning

With the recent advancements in fine-tuning transformer models on low-resource datasets, one technique that stands out is SetFit [23]. It was published in the paper "Efficient Few-Shot Learning Without Prompts," and it can be used to fine-tune Sentence Transformer models on small datasets.It uses the already-present world and language knowledge of pre-trained Sentence Transformers to learn useful representations from a limited number of labelled examples. According to the authors of the techniques, this type of fine-tuning can even work with 8 or 16 data points per class. Once the fine-tuning of the sentence transformer is completed, we can use it to generate embeddings for the entire dataset, which will then be used to train a classifier as shown in figure 18. Since our project also deals with biomedical datasets, which are notorious for their small size, we decided to investigate the potential of SetFit for classifying individuals with AD based on their Cookie Theft picture descriptions. From a theoretical point of view, SetFit is a promising approach to leverage the power of pre-trained language models. However, despite its theoretical appeal, our results from using SetFit were disappointing.

We experimented with various subsets of the training data, ranging from as few as ten examples to the entire training set. Despite our efforts, the best-performing SetFit model achieved an accuracy below 50% and an F1 score under 0.5. These results indicate that the model struggled to learn meaningful representations and discriminate between individuals with AD and cognitively normal controls.

Secondly, the pre-trained Sentence Transformer used in SetFit may not be well-suited for the specific domain of AD-related language data. The model's pre-training data encompasses a wide range of general language documents, which might not adequately represent the linguistic patterns observed in individuals with AD. Fine-tuning on a few examples from our specific domain may not be sufficient to adapt the model to capture these relevant features.

Lastly, the limited size of our dataset may have hindered SetFit's effectiveness. Even though SetFit is designed to work with small datasets, the complexity of our classification task and the intrinsic variability in language use across individuals may require more examples to train a more robust model.

4.1.6 Zero-Shot and Few-Shot Classification via LLMs

With the popularity of language models like Chatgpt and Claude-3, we decided to test their zero-shot and few-shot capabilities for our task. Specifically, we utilized the BioMixtral model [24], which is a pre-trained LLM specialized for biomedical and clinical text.

In the case of zero-shot classification, we directly use the pre-trained LLM to classify examples without giving it any additional fine-tuning or reference examples. The model is prompted with a description of the task using natural language and then given the examples that we need to classify. It then uses its pre-existing knowledge base to generate the predictions. On the other hand, few-shot classification involves providing the LLM with a small number of labelled examples (typically 1-5 per class) and using them to guide the model's predictions.

In both zero-shot and few-shot classifications, the initial prompt given to the model can have a noticeable effect on the quality of the predictions, so we tested a few variations. The one that seemed to work the best was the following: *"In this task, your goal is to predict if the speaker has Alzheimer's disease based on their description of the Cookie Theft picture from the Boston Diagnostic Aphasia Examination. In this test, the participant is shown a picture depicting a boy stealing cookies from a cookie jar while his sister and mother are distracted. The participant is asked to describe the picture in as much detail as possible. Note: This task is critical to me, so I need you to be accurate in your predictions."*

In the case of the few-shot classification, we also added two examples for each class. The note at the end of the prompt that states, "This task is critical to me, so I need you to be accurate in your predictions." is an engineering trick that seems to improve the accuracy of the final results by a few points.

Contrary to the promising results reported in the literature for zero-shot and few-shot classification using LLMs, our experiments with the BioMixtral model did not produce satisfactory results. The model struggled to accurately classify individuals with AD based on their Cookie Theft transcripts, even when provided with a few labelled examples. Our highest accuracy with this approach was around 70%, while the F1 Score was around 0.6.

There are a few reasons that can explain the less-than-ideal results of this approach. The main one is that the linguistic patterns associated with AD in the Cookie Theft transcripts may be too subtle and complex for the model to capture without explicit fine-tuning. While extensive, the pre-trained knowledge of BioMixtral may not be sufficient to distinguish between AD and non-AD individuals based solely on a few examples and a task description. This might have been further exacerbated by the possibility that transcriptions in our dataset may be significantly different from the text data used to pre-train BioMixtral. The data the model is trained on is likely biomedical research articles and clinical notes, which are pretty different from our dataset, even though they belong to the same domain. This type of mismatch might have been one factor limiting the model's ability to use its knowledge to our specific task effectively.

4.1.7 Wav2Vec2 Audio embeddings + logistic regression

The wav2vec2 model has shown significant promise in the field of speech processing by converting raw audio inputs into meaningful embeddings. Utilizing wav2vec for audio embeddings involves leveraging its capability to pre-train on large amounts of unlabeled audio data, capturing a rich representation of speech features. These embeddings can then be used as inputs for our classification task. In this approach, after extracting audio embeddings using wav2vec, we employ a logistic regression model to classify the presence of Alzheimer's. Logistic regression, a robust and interpretable machine learning algorithm, can effectively map these high-dimensional embeddings to binary outcomes, providing a probabilistic interpretation of the likelihood of Alzheimer's. This approach gave an accuracy around 60%.

4.1.8 Wav2Vec2-Conformers Audio embeddings + audio augmentations + logistic regression

We also experimented with Wav2Vec2-Conformers, which replaces the attention block with the conformer block in order to improve the model's ability to capture local dependencies in the data. In previous research, Wav2Vec2-Conformers has demonstrated better performance than vanilla Wav2Vec2 by lowering the word error rate for the same corpus. In this case, we introduce audio augmentations such as added noises and pitch transforms and then extract the hidden state of the final layer of Wav2Vec2-Conformers. Subsequently, given the limitations of only extracting mean pooling of the last hidden state, concatenate the mean pooling,

the min pooling, and the max pooling altogether. However, when providing the concatenated to a logistic regression, the accuracy and F1-scores underperform relative to text-only baselines and ended up with an accuracy and recall similar to the Wav2Vec2 approach.

4.2 Algorithms

- Main classification algorithm (with text input): preprocess text → add Bert summary → calculate Tfidf vector + extract Bert embedding → Logistic regression classifier.
- Main regression algorithm (with text input): preprocess text → calculate Tfidf vector + extract Bert embedding → ridge regression.

4.3 Implementations

- For classification and regression we use Scikit-Learn Logistic regression and Ridge regression library.
- For neural classification, we use Scikit-Learn MLP classifier
- For text embedding extraction we use S-Bert (sentence-transformer) library which has simple implementation and a variety of encode models to experiment.
- For audio transcription we use Whisper library which has intuitive and simple implementation and doesn't require big GPU power.
- For audio embedding extraction we use Wav2vec2 and similar libraries to process the audio files. It requires bigger GPU power and we often need to run them on cloud GPU such as the Colab environment.

5 Evaluation

5.1 Data splits

5.1.1 Original dataset:

For the original dataset, we have a total of 166 examples. We divided randomly them into 2 equal subsets (ratio 50/50) for training and development (evaluation), with a common random seed number 42. Each subset contains 83 examples.

5.1.2 Pitt dataset:

When splitting the imbalanced Pitt dataset, we recognized the need to conduct a stratified split to ensure that each split had similar proportions for the features of diagnosis, gender, and a binary variable of age before 65 or not for the first test of a participant. We also recognized that we needed to prevent leakage of data between the splits since a participant can have multiple tests over the period of many years. As a result, each participant can only be in one split. In the end, the train data has 174 participants corresponding to 337 examples, the dev dataset has 59 participants corresponding to 115 examples, and the test dataset has 59 participants corresponding to 97 examples.

5.2 Classification

5.2.1 Classical Baseline

Tfidf + Logistic Regression Classification: The classical baseline (Tfidf) models work surprising well with the following results 6:

It's also worth to note that, in the original dataset, the classification model is set with default settings (None) for "class_weights" parameter. However, in the Pitt dataset, because the dataset is less balanced on the distribution of AD / CN labels, the "class_weights" parameter is set to "balanced".

Dataset	Train accuracy	Train recall	Dev accuracy	Dev recall	Test accuracy	Test recall
Original	0.96	1.00	0.73	0.88	0.80	**0.97**
Pitt	0.88	0.86	0.75	0.72	**0.85**	0.85

Table 6: Classical baseline model performance

Dataset	Model	Train accuracy	Train recall	Dev accuracy	Dev recall	Test accuracy	Test recall
Original	Classical Tfidf	0.96	1.00	0.73	0.88	0.80	**0.97**
Original	Hybrid	0.92	0.96	0.76	0.85	**0.85**	0.94
Pitt	Classical	0.88	0.86	0.75	0.72	0.85	0.85
Pitt	Hybrid	0.91	0.90	0.79	0.77	**0.92**	**0.93**

Table 7: Classical baseline model performance

If we set class_weights parameter to None in the classification model for the Pitt dataset, the test scores are 0.88 and 0.96 for accuracy and AD recall respectively. However, because the Pitt dataset is not as balanced as the original dataset, we decided that setting class_weights = "balanced" is more appropriate.

We can see in the original dataset, the gap between train scores and test scores are larger than the Pitt dataset. It suggests that with more train data, the model are less likely to overfit on train set, and perform not as well on test set. It results in a better generalization of the model with Pitt dataset than with the original dataset.

However, the gap between development set scores and test scores are still relatively high. With the original dataset, development set accuracy and recall scores are (0.73, 0.88) and test set scores are (0.80, 0.97). With the Pitt dataset, the development scores are (0.75, 0.72) and test scores are (0.85, 0.85), suggesting there still can potentially be some limitation on generalization ability of the Classical Tfidf models.

5.2.2 Hybrid (joint) model

The hybrid architecture utilize both classical and neural approach into a simple, self-contained and efficient pipeline. First, it does an automated summarization of the text and add the summary to the main text. Second, it converts text to long, sparse vector (v1) using classical Tfidf vectorizer. Third, it convert the same text to shorter, dense vector (v2) using SBert text embedding tool with LaBSe embed model. Then, it combines (concatenate) v1 and v2 to create a longer vector v3 representating the text with its keywords and semantic meanings.

The hybrid models perform better than classical Tfidf baseline models in accuracy in both datasets (original and Pitt), as in the following table 7

It's worth to note that, the hybrid model for Pitt dataset are fine-tuned with hyper-parameter tuning process with various settings with Tfidf and Logistic regression components such as class weight, number of features, etc. The hyper-parameter tuning process for hybrid model for the original dataset has been done, but did not bring better outcome than manual hyper-parameters selected during model development.

As we can see, with the original dataset, the hybrid improved the accuracy on test set from 0.80 to 0.85, although the recall went slightly down from 0.97 to 0.94. Moreover, the gap between train accuracy (0.92) and test accuracy (0.85) is smaller than with classical model (0.96 and 0.80 respectively). It suggests better generalization ability of the hybrid model.

With the Pitt dataset, the hybrid model also improved the accuracy on test set from 0.85 to 0.92, and recall from 0.85 to 0.93. Similarly, the gap between train accuracy scores (0.91) and test scores (0.92) are narrower than the classical model (0.88) and (0.85) respectively.

There is still a gap between development scores (0.79, 0.77) compared to test scores (0.92, 0.93), suggesting there still a need for more data to improve generalization ability.

Dataset	Model	Train accuracy	Train recall	Dev accuracy	Dev recall	Test accuracy	Test recall
Original	Classical Tfidf	0.96	**1.00**	0.73	**0.88**	**0.80**	**0.97**
Original	Neural classifier	**1.00**	**1.00**	**0.84**	0.85	0.70	0.71

Table 8: Classical vs Neural, Original dataset

Dataset	Model	Train accuracy	Train recall	Dev accuracy	Dev recall	Test accuracy	Test recall
Pitt	Classical Tfidf	0.88	0.86	**0.75**	0.72	**0.85**	**0.85**
Pitt	Neural Classifier	**1.00**	**1.00**	0.73	**0.75**	0.72	**0.85**

Table 9: Classical vs Neural, Pitt dataset

5.2.3 Neural classification model

The neural classification model utilizes SBERT text embedding tool with LaBSe embedding model to convert text into fixed-length vectors, and run classification training and prediction using Sk-learn MLP (Multi-layer Perceptron) Classifier. The neural classifier has 3 hidden layers of size 500 neurons each.

With the original dataset, the scores are in the following table 8

We can see the Neural model clearly overfitted on the train set, with train scores of 1.00 on both accuracy and recall. It did much worse on the test set, with accuracy and recall are 0.70 and 0.71 respectively. The neural model performance is lower than the classical Tfidf model, with test scores on accuracy and recall are 0.80 and 0.97 respectively.

With the Pitt dataset, the scores are in the following table 9: Similarly, with the Pitt dataset, the Neural model overfitted on the train set, with both accuracy and recall scores are 1.00. It did worse with test scores are 0.72 (accuracy) and 0.85 (recall). Compared with the classical Tfidf model, the classical model did better on the test set with accuracy and recall scores of 0.85 and 0.85 respectively. Also the gap between train scores and test scores in Neural model are bigger than classical Tfidf model.

Our best explanation about why Neural model did worse than the classical Tfidf model for our task is likely because of limited size of the dataset. For original dataset, the train set count is 83 examples, and on the Pitt set, it is 337 examples. Neural networks tends to do well with massive number of examples to train on, and tends to be unstable when trained with small datasets[25]. Therefore, in low-resource situations like the task we are working on, with very limited data, neural (deep learning) model does not seem to have advantage over classical model such as Tfidf and Logistic regression classifier.

When there are much more data available for the Alzheimer classification task, we think neural model might do much better.

5.3 Regression

Table 10: RMSE of Classical and Hybrid Models on Different Datasets

Dataset	Model	Train RMSE	Dev RMSE	Test RMSE
Original	Classical	3.33	5.50	4.76
Original	Hybrid	0.78	5.33	**4.49**
Pitt	Classical	2.78	4.57	**5.33**
Pitt	Hybrid	0.51	4.56	5.47

For the regression task, we first find that generalizability is a challenge as both the classical and hybrid models overfit on the respective training data. Nevertheless, the classical approach performs surprisingly well on the test dataset. While it significantly underperforms the hybrid approach for the ADReSSo dataset for

the training dataset, we observe that it only lags behind slightly for the dev and test datasets. This indicates that the hybrid approach is severely overfitting the data, which is even more apparent when evaluating the Pitt dataset. The hybrid approach's RMSE for the training dataset once again is drastically lower than the classical dataset but we find that the two approaches are on par with each other for the dev dataset and that the classical approach actually achieves superior results for the test dataset.

5.4 Hyperparameter tuning

For the hybrid model with the Pitt dataset, we have done grid search on the following parameters for the Tfidf vectorizer and Logistic regression classifier:

- **tfidf_params_max_features** = [None, 1000, 5000, 10000] that set the maximum of words to be used in Tfidf vectorizer
- **tfidf_ngrams** = [(1, 1), (1, 2), (1, 3), (1, 4), (2, 3), (2, 4)] that set the range of ngram to use in Tfidf vectorizer
- **lr_class_weights** = [None, "balanced", "AD":0.65, "CN":0.35, "AD":0.4, "CN":0.6, "AD":0.45, "CN":0.55, "AD":0.5, "CN":0.5, "AD":0.55, "CN":0.45, "AD":0.6, "CN":0.4, "AD":0.65, "CN":0.35] to set the weights on the labels (AD/CN) on the logistic regression classifier.

For each combination of these 3 parameters, we trained on the train set, and did evaluation on the development set to select the best model with the highest average of overall score and AD F-1 score. We chose F-1 score for hyperparameter tuning because it would bring a more balanced model that did relatively well on all scores, and to prevent overfitting on recall score alone.

The precision, recall and F-1 scores are defined as followed[17]:

$$Precision = \frac{True\ Positives}{True\ positives + False\ Positives}$$

$$Recall = \frac{True\ Positives}{True\ Positive + False\ Negatives}$$

$$F_1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

After running through the exhaustive grid search, the best hyperparameter for use in hybrid model with Pitt dataset are:

- **Tfidf max_feature = 10,000**, meaning using maximum 10,000 words
- **Tfidf ngram range = (1, 2)**, meaning considering unigram and bigrams
- **Logistic regression class_weights = "balanced"**, meaning it automatically balance weights based on the number of AD and CN examples in the train set.

It's worth to note that the fine-tuning process can result in slightly different results on different OSs and environment. For consistency, we recommend manually setting these parameters for the final model.

6 Analysis

6.1 Results vs expectations

We have attempted many approaches to solve the task, and we put slightly greater part of our efforts on the classification task (task1), because we believe it is more practical in real world use to detect signs of Alzheimer's Disease as early as possible. Some of the approaches are more relatively successful than the others, and we would summarize and highlight the notable scores in the following table11, with linguistic-feature baseline scores [26] provided by the organizer, using LDA classifier and linguistics features 11:

Dataset	Model	Train accuracy	Train recall	Dev accuracy	Dev recall	Test accuracy	Test recall
Original	Organizer baseline	-	-	-	-	0.75	0.62
Original	Classical Tfidf	0.96	**1.00**	0.73	**0.88**	0.80	**0.97**
Original	Hybrid	0.92	0.96	0.76	0.85	**0.85**	0.94
Original	Neural classifier	**1.00**	**1.00**	**0.84**	0.85	0.70	0.71
Pitt	Classical	0.88	0.86	0.75	0.72	0.85	0.85
Pitt	Hybrid	0.91	0.90	0.79	0.77	**0.92**	**0.93**
Pitt	Neural Classifier	**1.00**	**1.00**	0.73	0.75	0.72	0.85

Table 11: Overall classification models summary

For the original ADReSS dataset, comparing with the organizer baseline scores (accuracy 0.75 and AD recall 0.62) for the classification task, both our Classical (Tfidf) and Hybrid outperformed it considerably on both accuracy and recall scores, especially on AD recall scores. Out of the 2 models, the Hybrid model scores better on accuracy (0.85 vs 0.80) but lower on AD recall (0.94 vs 0.97). For the regression task, both our baseline and hybrid models' RMSE values of 4.76 and 4.49 also outperformed the baseline RMSE scores, which were 5.20. It suggests that Tfidf, which focuses on the keyword frequency of each text / document, recognize the examples associated with AD label better. However, for overall accuracy, the hybrid model, which combined keyword frequency as well as semantic meaning of the text, is better. It suggests that when considering multiple factors (keyword and semantic meaning), the Hybrid model is better at classifying both labels (AD and CN).

For the Pitt dataset, because we split the Pitt dataset into train, development and test set using specific random seeds, so our test set can be different from others' works. Therefore it might not possible to directly compare with other papers. For example, using BERT-large embeddings and Logistic regression, Roshanzamir et al. [27] achieved accuracy of 0.88 and recall of 0.84 on the Pitt dataset. However, we are refrained from directly comparing with this result for both the classification and regression task.

Within our models for the classification task, the Hybrid model is the best performing one, with accuracy and recall are 0.92 and 0.93 respectively, comparing to the Classical (Tfidf) model (0.85 and 0.85 respectively). The gap between train scores (0.91, 0.90) and test scores (0.92, 0.93) is also narrower than with the original dataset train and test scores are (0.92, 0.96) and (0.85, 0.94) respectively.It suggests that the Hybrid approach has benefited from more data in the Pitt dataset than the original dataset. For the regression task, it is not as clear whether the hybrid model outperforms the classical model and deserves additional investigation going forward.

In both datasets, the Neural method which used Multi-layer Perception Classifier instead of Logistic regression, performed worse than Classical and Hybrid model. It suggests that there might not be enough data to make use of deep learning / neural method advantages. We might need much more data, probably in the thousands or more examples, to better utilize neural classifier.

In both datasets, the Classical (Tfidf) model performed surprisingly well, exceeding the organizer's baseline scores. It suggests that the focus on keyword frequency does relatively well for the classification task.

6.2 Model reliability

Although we have tried our best to increase the size of dataset by using the Pitt dataset (separately from the original ADReSS dataset), the size is still relatively very limited. It leads to limited generalization ability of the model, which can be improved by getting much more data.

The text transcriptions from the original dataset and the Pitt dataset are likely to be processed automatically by some ASR models rather than manually transcribed, so there could be inaccuracies in the text transcription. In the Pitt dataset, we also selected speech by the participants / patients only, and ignored

the text by investigator or staff members. If the model is to be evaluated with extra dataset, the audio or transcriptions should also remove audio segments or transcript segments by non-participants (e.g. staff members, doctors, etc) to make the results more consistent.

Another possible limitation is there are some randomness involved. We splitted the datasets into train, development and test sets using specific seed number (42), and sorting the filenames when reading text files. However, with the limited size of the datasets, the scores can vary significantly with different seed numbers.

Although we have set specific versions for the important packages we used (sentence-transformers==2.7.0 and bert-extractive-summarizer==0.10.1), some variations can occurs because of difference in Operating Systems, sub-packages or dependencies versions, encoding model versions, etc that are beyond our control.

6.3 Model application

The models (Classical and Hybrid) might be further tested and fine-tuned with more real-world data such as in closed clinical settings. However, we suggests to have audio and transcriptions formatted similarly to how we processed the data. For example, in the Pitt dataset, we only selected speech by the participants and removed speech / text by investigators or staff.

The model also can benefit with running on a capable computer with at least 6GB of GPU in order to run BERT embedding and summarization models.

7 Future Work

7.1 Shortcomings

The most critical shortcoming of our analysis is the limited size of our dataset that undermines the generalizability of the model for different settings in real life situations. While we explored a number of additional datasets, we found that data incompatibility and lack of transparency made it challenging to merge them together. In addition, our dataset is primarily from one source and are all collected in English from participants in the United States. Consequently, the performance of the model may differ drastically depending on how the randomized splits come out.

Secondly, many of the deep learning models are difficult to interpret and understand. As interpretability is one of the most critical components of model building in the medical field, our traditional machine learning techniques can be explained but that is not the case with our neural models. It is difficult to explain why certain models work well but others do not.

7.2 Additional data

The most crucial element of future work is to collect additional validation text and audio data that can be used to both train and evaluate the model. As this is a common challenge in the medical domain, we recognize that this requires a substantial investment in time and energy from our partner. However, we believe that this is more critical than building better models or building an application as the model cannot be wholly trusted until we have additional data sources to evaluate the reliability of the model.

7.3 Deployment

In future iterations of this project, we aim to develop a user-friendly web interface using Gradio to allow easy uploading and processing of multiple audio files. The interface will allow users to select and upload Cookie Theft test audio recordings. Once uploaded, the audio files will be automatically transcribed using OpenAI's Whisper model. The transcriptions obtained from the Whisper model will then be passed to our best-performing machine learning model, which has been trained to classify individuals with Alzheimer's disease based on their linguistic patterns. The model will process each transcription and generate a prediction indicating the likelihood of the speaker having Alzheimer's disease. The predictions, their transcriptions, and filenames will be compiled into a CSV file for easy analysis. Users can view the results directly within the web interface or download the CSV file for further examination. By deploying our model through a Gradio web interface, we aim to make it more accessible and user-friendly for researchers to utilize our system for

the early detection of Alzheimer's disease. The easy-to-use interface, coupled with the ability to process multiple audio files simultaneously, will streamline the analysis process and enable users to obtain results quickly and efficiently.

7.4 Multi-lingual model

We would like to build a multi-lingual model that would work on different language inputs for text and audio. We know that DementiaBank does contain datasets in different languages and build models based on these standardized datasets. In addition, we can integrate our existing models with multi-lingual large language models to conduct classification of texts in multiple languages and better understand how the deterioration of cognitive abilities is reflected differently across different languages.

8 Conclusion

We have the task of detecting potential Alzheimer's Disease from a person's spontaneous speech about a specific Cookie Thief picture. We worked on both original (ADReSS) and Pitt datasets, and mainly on text transcription data.

Using Classical model, which uses Tfidf vectorizer and logistic regression classifier, we achieved accuracy and AD recall of 0.80 and 0.97 respectively on the ADReSS dataset, and 0.85 and 0.85 on the Pitt dataset for the classification problem. For the regression task, our ridge model achieved RMSE values of 4.76 and 5.33 on the ADReSS and Pitt dataset respectively.

Using Hybrid model, which uses Tfidf vectorizer, Bert summarizer and LaBSe Bert sentence embeddings and Logistic regression, we achieved accuracy and AD recall of 0.85 and 0.94 respectively for the original (ADReSS) dataset, and 0.92 and 0.93 for the Pitt dataset. For the regression task, the ridge model had RMSE scores of 4.49 and 5.47 on the ADReSS and Pitt dataset respectively.

Although the gap between train scores (0.91, 0.90) and test scores (0.92, 0.93) are narrow with Pitt dataset and Hybrid model, there is still some gap between development scores (0.79, 0.77) and test scores (0.92, 0.93), which is also reflected in the regression task as well. It suggests that although the generalization has improved, it still can benefit from having more data for training and testing.

8.1 Project management

In terms of capstone project management, we have learnt how to use knowledge and skills in our program into solving real world problems. We tried new methods and techniques such as how to work with audio data, for example doing automatic speech recognition or extract audio embeddings, etc.

We also get to know some challenges in medical domains, such as limited available data sources, privacy and ethics issues. We also get familiar with latest research papers about dementia and Alzheimer's Diseases, that we can learn a lot from.

One of the most difficult challenges is the low-resource settings we are working with. There is very little data to begin with, and although we got more data, it is still limited comparing to what we are used to do. There are some overlaps between the ADReSS and Pitt datasets. Although we tried to detect and remove the overlaps, but ultimately we decided to run the models separately with each dataset, to prevent leaks between them.

9 Current Code Execution Instructions

In order to run our code, please follow the following instruction:

1. Install standard data science environment (Python, Numpy, Pandas, Jupyter Notebook, etc)
2. Install SBert text embeddings and Bert summarizer libraries with specific versions as follow:
   (a) **pip install sentence-transformers==2.7.0**
   (b) **pip install bert-extractive-summarizer==0.10.1**

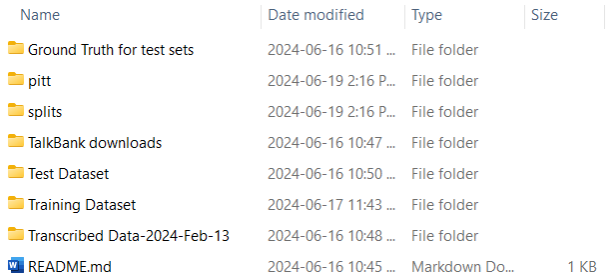


Figure 19: data folder structure

3. Clone our repo by
   (a) **git clone https://github.com/M-I-Dx/alzheimers-detection.git**
4. Download DementiaBank ADReSS dataset and Pitt dataset, place them in the **data** folder with structure the same as the illustration figure 19
5. Run data preprocessing code:
   (a) Original dataset:
      i. **/src/data_preprocessing/original/01_process_test_set.ipynb**
      ii.
   (b) Pitt dataset:
      i. **/src/data_preprocessing/pitt/00_split_data.ipynb**
      ii. **/src/data_preprocessing/pitt/01_preprocessing_pitt.ipynb**
      iii. **/src/data_preprocessing/pitt/02_data_splits_pitt.ipynb**
6. Run the EDA code:
   (a) Original dataset
      i. **/src/EDA/original/01_EDA_data_splits.ipynb**
   (b) Pitt dataset
      i. **/src/EDA/pitt/01_eda_pitt.ipynb**
7. Classical (Tfidf) baseline model:
   (a) Original dataset:
      i. **/src/classical_baseline/original/01_classical_baseline_original.ipynb**
   (b) Pitt dataset:
      i. **/src/classical_baseline/pitt/01_classical_baseline_Pitt.ipynb**
8. Neural classifier model:
   (a) Original dataset:
      i. **/src/neural_classification/original/01_neural_classification_original.ipynb**
   (b) Pitt dataset:
      i. **/src/neural_classification/pitt/01_neural_classification_pitt.ipynb**
9. Hybrid model
   (a) Original dataset:

Week number	Start and end dates	Tasks to be done
1	April 30 - May 5	Exploratory Data Analysis, Meeting with Dr. Kenneth Madden (Capstone partner) (All members)
2	May 6 - May 12	Project planning EDA (continue) - all members Baseline models Classical (Minh + Qihan) Text Neural (Minh + Qihan) Text + Audio (Mandeep) + Audio EDA (Daniel)
3	May 13 - May 19	Feature engineering (all members)
4	May 20 - May 26	Feature engineering (all members) Fine-tuning models (all members) LLM-based approach (Minh)
5	May 27 - June 2	Fine-tuning models (all members) LLM-based approach (Minh)
6	June 3 - June 9	Final model(s) selection (all members)
7	June 10 - June 16	Wrapping up, code cleaning and organization (all members)
8	June 17 - June 23	Report writing + Presentation preparation (all members)
9	June 24 - June 30	Presentation + Delivery (all members)

Table 12: Original timeline

i. **/src/hybrid/original/01_hybrid_original.ipynb**

(b) Pitt dataset:

i. **/src/hybrid/pitt/01_hybrid_pitt.ipynb**

10. Evaluation code (for original dataset)

(a) go to **/src/evaluation/** folder

(b) run **python evaluate.py task1_gold.csv test_results_task1_classical.csv** for classical model evaluation

(c) run **python evaluate.py task1_gold.csv test_results_task1_hybrid.csv** for hybrid model evaluation

(d) run **python evaluate.py task1_gold.csv test_results_task1_neural.csv** for neural classifier model evaluation

10 Timeline

We believe that the project is successful in terms of outcome, project management and achieving the overall objectives, with relatively equal contributions of all members. Our main principle is volunteer-based, which means each member can actively volunteer to work on the tasks that one is most familiar with and interested in. We also tried to work in parallel, so one member doesn't need to wait for another member to proceed. We shared our updates and discussed regularly on Slack group channel and in weekly internal meetings.

We summarized our initial project planning 12 as well as actual timeline 13 . The project planning went mostly consistent with the original timeline, except in the last few weeks, we put efforts in solving the data overlaps issues, and decided to re-run some selected models on the Pitt dataset separately from the original (ADReSS) dataset to avoid data leak.

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Week number	Start and end dates	Tasks done
1	April 30 - May 5	**Meeting with Capstone partner, obtain dataset.** Minh: text data preprocessing, text EDA, text baseline models.
2	May 6 - May 12	**Project planning, regression task, code repo setup.** Daniel: Audio EDA. Minh: hybrid models with text embeddings. Qihan: conduct EDA. Mandeep: Whisper + Transformer models
3	May 13 - May 19	**Working with additional datasets.** Daniel: Audio Embedding model. Minh: tried baseline and hybrid models with new data. Experimented with SetFit finetuning library. Qihan: review new datasets. Mandeep: Reviewed the WLS dataset
4	May 20 - May 26	**Fine-tuning models. Found possible overlaps between Pitt and ADReSS datasets.** Daniel: Audio Embedding model. Minh: worked on hybrid models with new data. Experimented with larger embed models (e.g. Mistral-E5). Qihan: mmse preprocessing and implement regression model. Mandeep: Explored the SetFit tinetuing on the ADReSS dataset
5	May 27 - June 2	**Data cleaning to remove overlaps.** Minh: experimented with text-embeddings based similarity to detect overlaps. Experimented with Bert Summarizer. Qihan: start report skeleton, provide progress update to Ken, and experiment with wav2vec2-conformer models. Mandeep: Dynamic Time Wrapping and other audio similarity detection techniques to detect overlap
6	June 3 - June 9	**Data augmentation and more fine-tuning. Discussing about doing the Pitt dataset separately from ADReSS dataset to avoid data leak.** Minh: worked with LLM-based text paraphrasing to generate additional text data. Rework models on Pitt dataset separately from original dataset. Qihan: continue experimenting with wav2vec2-conformer models with data augmentation. Mandeep: Started working on the deployment code + hybrid audio and text embeddings for the classification task
7	June 10 - June 16	**Final report draft and repo organization.** Daniel: help with report. Minh: cleaning code for merging to main. Worked on baseline and hybrid models on Pitt dataset. Worked on final report draft. Qihan: submit code to main and review pull requests. Mandeep: Completed the deployment code and worked on the report
8	June 17 - June 23	**Final report, partner presentation.** Daniel: final report Minh: worked on final report and presentation slides. Creating PR to merge code to main branch. Qihan: Work on final report and presentation. Gave the presentation to Ken. Mandeep: Worked on the final report. Gave the presentation to Ken

Table 13: Actual timeline

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